Change Report - Stable Pick-and-Place + Sensor Integration

4-DOF + gripper robotic arm (6-DOF frame; wrist roll deferred)

Repository: https://github.com/saumyadebug/robotic-arm-pick-and-place

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1. Purpose

Bring the heavy metal arm to a stable pick-and-place cycle, then integrate the HC-SR04 ultrasonic sensor and TCS3200 colour sensor for Red/Green/Blue sorting. Yellow sorting, wrist roll and inverse kinematics remain deferred.

2. Motion Fix

The previous firmware used the shoulder for the deep reach. That hung the whole arm in front of the base, the shoulder could not hold it, and the arm collapsed and could not lift.

New strategy: the shoulder stays vertical (260/258). The elbow performs the reach (300 -> 278 -> 248 -> 218). Lifting is staged in reverse (218 -> 240 -> 262 -> 284 -> 300), so only the light forearm moves first. The base rotates only after the arm is folded upright.

Operational clamps: base 335-415, shoulder 250-275, elbow 205-310, wrist 370-380, gripper 120 open / 200 closed. Bin angles: RED 335, GREEN 375, BLUE 415.

3. Firmware Changes

code/pick_and_place/pick_and_place.ino

- Elbow-only reach, staged lift, base rotation only while upright, gripper left open when rotating home.

code/integrated_sorting/integrated_sorting.ino

- Rewritten FSM with the stable motion, ultrasonic debounce, TCS3200 R/G/B/C scan, colour retries and matching bins.

code/ultrasonic_test/ultrasonic_test.ino

- HC-SR04 test with the 2-10 cm pick-zone window and correct wiring notes.

code/color_sensor_test/color_sensor_test.ino

- TCS3200 test with 3-sample averaging, clear-channel reading and relative classification.

code/end_effector_test/end_effector_test.ino

- Corrected to channel 7, open 120 / closed 200.

code/servo_testing/servo_testing.ino

- Safe small-range test that skips faulty channel 3 and dead channel 4.

code/joint_calibration/joint_calibration.ino

- Calibrated range table; ignores faulty/dead channels.

4. Sensor Wiring Added

- HC-SR04: VCC=5V, GND=GND, TRIG=D9, ECHO=D10.
- TCS3200: VCC=5V, GND=GND, S0=D4, S1=D5, S2=D6, S3=D7, OUT=D8, OE=GND.
- Sensors are powered from Arduino 5V; servos stay on the external 5-6 V buck converter.
- All grounds are common with the Arduino and PCA9685.
- Guide: docs/sensor_integration.md; pin map: hardware/pin_connections.md.

5. Documentation Updated

- README.md - current scope, progress, servo map, sensor wiring and bring-up order.
- docs/safe_motion_chart.md / .pdf - current calibrated values and elbow-reach sequence.
- docs/sensor_integration.md - step-by-step wiring, tests and colour calibration.
- docs/testing_procedure.md - full commissioning checklist.
- docs/troubleshooting.md - collapse, sensor and power faults.
- docs/servo_calibration.md, docs/motion_control.md, docs/working_principle.md, docs/system_architecture.md - aligned with current hardware.
- hardware/pin_connections.md, hardware/power_supply.md, hardware/components.md - corrected channels and sensor power.

- tests/* - servo calibration values recorded; integration results marked pending instead of claiming unverified passes.
- IMPLEMENTATION_PLAN.md and docs/implementation_plan.md / .pdf - current plan and acceptance criteria.

6. Known Hardware Faults

- PCA9685 channel 3 has no output - never use it.
- Wrist-rotation servo on channel 4 holds but never rotates, even with a bare shaft - replace it.
- Gripper only has useful travel from 120 to 200 ticks.
- The shoulder cannot safely perform a deep reach; keep it near vertical.

7. Next Steps

- Wire the ultrasonic and colour sensors as shown in docs/sensor_integration.md.
- Run the ultrasonic test and confirm the 2-10 cm window.
- Run the colour test and tune DARK_LIMIT / MARGIN_PCT for the actual lighting.
- Run the integrated sorting sketch with the base clamped and the arm supported during the 6-second startup.
- Record real results in tests/ before marking any cycle as PASS.